Introduction to Differential Geometry

Professor Alan Huckleberry

Jacobs Univestiy Bremen

Semester: Spring 2019

Friday, 15. February 2019

Schedule for the semester

We have agreed that the lectures will take place during the lunch break on Thursdays, starting at shortly before 13:00 so that there is time to get something to eat, at the regular times on Friday, i.e., the slots at 9:45 and 15:45, plus an additional slot after the late afternoon lecture on Fridays, i.e., the slot at 17:15. More often than not I will stay over on Friday night so that we have Friday evening and Saturday morning for continued discussions.

The following is a detailed list of the lectures (date and number of lectures).

	Date	No. of Lectures
February	Friday, 08. February 2019	Two lectures
	Thursday, 14. February 2019	One lectures
	Friday, 15. February 2019	Three lectures
March	Friday, 15. March 2019	Three lectures
	Thursday, 21. March 2019	One lecture
	Friday, 22. March 2019	Three lectures
	Thursday, 28. March 2019	One lecture
	Friday, 29. March 2019	Three lectures
April	Thursday, 04. April 2019	One lecture
	Friday, 05. April 2019	Three lectures
May	Thursday, 16. May 2019	One lecture
	Friday, 22. May 2019	Three lectures
	Reading period	One lecture

Above we have listed a total of 26 lectures. Since the semester lasts 13 weeks, we have therefore planned lectures which would correspond to 2 lectures per week, i.e., the standard number of lectures.

Introduction to Differential Geometry

Professor Alan Huckleberry

Jacobs Univestiy Bremen

Semester: Spring 2019

Sunday, 10. February 2019

Homework system

Try the following system for the first week of the semester, i.e., as homework for Lectures 1 and 2. Since the system is a bit new, everyone who gives it a try for this first week will get full credit.

Homework system

In any endeavor where the goal is to understand something, the most important first steps are
1. Being able to pose good "first" questions.
2. Being able to outline at least an initial approach to answering
the first questions with the hope of leading to at least a certain degree of understanding. Each (standard) lecture block consists of four lectures which take place on Thursdays and Friday. After the Friday lectures I will post a summary of the lectures in the block. This summary (sketch) should be used as a guide for 1. and 2. . In free time around and following the blocks the students will have a chance to chat with Latif and AH about all of this. Then by Sunday evening they should send files (pdf) to Latif which pose the good first questions, i.e., 1., and the initial approaches to answering them, i.e., 2., of the previous block of lectures of one week before. As an example, the files on the lectures of last Friday (Feb. 8) will be due on Sunday, (Feb. 17).

Example.

1. Do I really understand the smooth structure on the tangent bundle?
2. Write down the isomorphism to the trivial bundle which is given by a chart and conceptually check that change of coordinates is linear in the fiber and depends smoothly on the base.

The output of this, i.e., that which is sent to Latif, should be roughly one page.

Grading: Latif will send back the files with comments. Clearly, there is no right or wrong with this kind of challenge. However, after the student has some experience with feedback from Latif, it should be clear if he/she is serious about this. A student who submits all of these papers $\pm\varepsilon$, and they indicate that he/she is serious about the matter, gets 100% for the grade on the homework. Extra credit will be given for those who go beyond 1. and 2.

Introduction to Differential Geometry

Professor Alan Huckleberry

Jacobs Univestiy Bremen

Semester: Spring 2019

Friday, 08. February 2019

Lectures: 1–2

In general the notes which accompany the videos will provide summary sketches of the contents of the lectures. Students should derive their own homework exercises from theses sketches.

Derivations

Let M be a manifold, p be a point of M and denote the image in $\mathbb{R}^n$ of a coorinate mapping of a neighborhood of p by U. Let $x_1, \ldots, x_n$ be the resulting coordinates and $\partial_1, \ldots, \partial_n$ the associated derivations on the algebra $C^\infty(U)$ of smooth functions. They form a basis of $Der(C^\infty(U))$ as a $C^\infty(U)$-module.

Tangent spaces and Tangent bundle

The tangent space T_pM can be defined via (equivalence classes of) curves or by evaluating elements of $Der(C^\infty(U))$ at p. In any case it is an n-dimensional vector space. As a set the tangent bundle of M is just the disjoint union $TM = \dot{\cup}_{p\in M} T_p$. It is equipped with its canonical projection π which maps $v_p \in T_pM$ to p. We may restrict it to U to obtain the bundle $\pi : TU \to U$. By evaluation at any p the above partial derivative operators form a basis of the vector space T_pM and therefore define a bundle isomorphism (linear in each fiber) to the trivial bundle $U \times \mathbb{R}^n$. Using the smooth structure of this trivial bundle we equip $TU \to U$ with a smooth vector bundle structure which, using a covering of M by coordinate patches, defines a smooth bundle structure on $TM \to M$. One checks that this structure is defined independent of the choice of the coordinates.

Vector fields

In general a section of a smooth fiber bundle $\pi : E \to B$ is a smooth map $s : B \to E$ with $\pi \circ s = Id_B$. In the case of the tangent bundle we refer to a section as a vector field. The space of such sections is denoted by $\Gamma(M, TM)$ or equivalently $Vect(M)$. This space is a $C^\infty(U)$-module and it acts on $C^\infty(U)$ by derivations, i.e., by $s(f)(x) = s(x)f$. In fact this establishes an $C^\infty(U)$-module isomorphism $\Gamma(U, TU) \to Der(C^\infty(U))$ or more generally $\Gamma(M, TM) \to Der(C^\infty(M))$.

Derivations on $\Gamma(E, B)$

The special feature of the tangent bundle is that its fibers have vector space structure which is compatible with the smooth structure. Now let $\pi : E \to B$ be any vector bundle and notice that its space of sections is a $C^\infty(B)$-module. So we ask about the derivations in $Der(\Gamma(E, B))$, i.e., mappings $D : \Gamma(E, B) \to \Gamma(E, B)$ with

$$D(fs) = fDs + D(f)s$$

where $f \mapsto D(f)$ is a derivation of functions, i.e., the application of a vector field to f. In that case on usually denotes $D(f)$ by $X(f)$, where X is the vector field in question and to emphasize that D depends on X one uses the classical notation ∇_X. Then

$$\nabla_X(fs) = f\nabla_X s + X(f).s\,.$$

This is how one differentiates sections of a vector bundle, in particular of the tangent bundle. We will discuss maps of the type ∇ in detail in the next lectures. Here we just close by remarking that the *connection* ∇ is an added structure on M which in particular will allow us to *parallel transport* tangent vectors along curves from one fiber to another and to define *curvature*. A given connection may or may not be appropriate for a given *application*. There are plenty of connections (an infinite dimensional space). Often the key question is whether or not there is a *good* connection and if so, how do we find it?!

Introduction to Differential Geometry

Professor Alan Huckleberry

Jacobs Univestiy Bremen

Semester: Spring 2019

Friday, 15. February 2019

Lectures: 3–6

Parts of these lectures were devoted to review and/or providing more concrete detail to the previous introductory lectures.

Manifolds

Here we are dealing with paracompact Hausdorff spaces M which are equipped with charts $\{(U_\alpha, \varphi_\alpha)\}$ which define the smooth structure on M. The homeomorphisms $\varphi_\alpha : U_\alpha \to V_\alpha \subset \mathbb{R}^n$ are C^∞-compatible in the sense that the chart-change maps

$$\varphi_{\alpha\beta} : V_{\alpha\beta} \to V_{\beta\alpha}$$

are smooth. This structure allows us to discuss the *commutative ring* $C^\infty(U)$ of smooth functions for any open set U in M.

Derivations and connection formalism

Let R be a ring such as those above. A *derivation* δ on R is a map $\delta : R \to R$ which preserves the additive structure of R and, with respect to the multiplicative structure, satisfies the Leibniz rule

$$\delta(fg) = \delta(f)g + f\delta(g)\,.$$

If $f \in R$ and δ is such a derivation, then $f\delta$ is a derivation and consequently the space $\mathrm{Der}(R)$ of all such derivations is an R-module. The main example for us is the space $\mathrm{Der}(R) = \mathrm{Vect}(M)$ of vector fields which is an $R = C^\infty(M)$-module.

If $\mathcal{M}$ is an R-module, then a first order *differential operator* on $\mathcal{M}$ is a pair (D, δ) where $D : \mathcal{M} \to \mathcal{M}$ preserves the additive structure, $\delta \in \mathrm{Der}(R)$ and the Leibniz rule

$$D(rm) = rD(m) + \delta(r)m$$

is satisfied. The space of such operators is denoted by $\mathrm{Diff}_1(\mathcal{M}, \mathcal{M})$. A *connection* in this setting is an R-module map

$$\nabla : \mathrm{Der}(R) \to \mathrm{Diff}_1(\mathcal{M}, \mathcal{M}),\ \delta \mapsto (\nabla_\delta, \delta),$$

satisfying the Leibniz rule $\nabla_\delta(rm) = r\nabla_\delta(m) + \delta(r)m$. In other words, a conection defines an action of $\mathrm{Der}(R)$ by first order differential operators on $\mathcal{M}$. Recall that in our manifold context $R = C^\infty(M)$ (or $C^\infty(U)$ for an open subset U of M) and $\mathrm{Der}(R)$ is the R-module $\mathrm{Vect}(M)$.

The key modules of differential geometry

The modules of relevance for us are the spaces $\Gamma(M, E)$ of sections of smooth vector bundles $\pi : E \to M$ on a smooth manifold M, the first example being the tangent bundle TM. Recall that the *entire* space E is the disjoint union $\dot{\cup} E_x$ of vector spaces (in our case over $\mathbb{R}$) which has a smooth manifold structure with the property that π is a maximal rank surjective smooth map. The smooth vector bundle structure is given by C^∞-compatible *frames* $\boldsymbol{e}$ on a family of open subset sets U which cover M. Such a frame is denoted by $\boldsymbol{e} = (e_1, \ldots, e_n)$ where $e_i : U \to M$ is a smooth map with $e_i(x) \in E_x$ and such that $(e_1(x), \ldots, e_n(x))$ is a basis of E_x. Every $v_x \in E_x$ is uniquely described by its coefficients c in this basis and therefore we have a map $v_x \mapsto (x, c(v_x)$ which defines a diffeomorphism $TU \to U \times \mathbb{R}^n$ which is linear in the fibers of E. Thus frames give the smooth vector bundle structure of E.

A *section* s of $E \to M$ is a smooth map $s : E \to M$ with $\pi \circ s = Id_M$. Note that if f is a smooth function on M and s is a section, then fs is again a section. Hence, the space $\Gamma(M, E)$ is a module over the ring $C^\infty(M)$ of smooth functions on M. A connection on E is then that which has been formalized above: $R = C^\infty(M)$, $\mathrm{Der}(R) = \mathrm{Vect}(M)$ and $\mathcal{M} = \Gamma(M, E)$. For X a vector field we have

$$\nabla_X(fs) = f\nabla_X(s) + X(f)s\,.$$

Connection matrix, Christoffel symbols

If $\boldsymbol{e} = (e_i, \ldots, e_n)$ is a frame of E over an open subset of M and ∇ is a connection on E, then in order to know how a vector field X acts on a general section $s = f_i e_i$ it is enough to know how it acts on the e_i. Hence consider

$$\nabla_X(e_i) = \omega_{ij}(X)e_j$$

where $\omega = (\omega_{ij})$ is a matrix of 1-forms on the open set being considered. Given a covering of the base manifold M with open sets U and frames for E, denoted by $\boldsymbol{e}$, over U, the associated connection matrices ω determine the connection.

In the special case where $E = TM$ is the tangent bundle, the form ω can be given by the associated Christoffel symbols. For this we have the following notation: Let U be a coordinate patch of M with associated coordinates $(x_1, \ldots, x_n)$. Here $\partial_i := \frac{\partial}{\partial x_i}$. Since every $X \in \mathrm{Vect}(U)$ can be described in the frame $(\partial_1, \ldots, \partial_n)$, in order to describe the connection we must only know $\nabla_{\partial_i}(\partial_j)$ which is just a linear combination (with coefficients being functions) of the frame elements:

$$\nabla_{\partial_i}(\partial_j) =: \Gamma^k_{ij}\partial_k\,.$$

The coefficient functions are known as the *Christoffel symbols* of ∇ in the frame $\boldsymbol{e}$.

Parallel transport

Here we first consider smooth curves γ contained in a coordinate neighborhood of a base point x_0. We let $I = [0, 1]$, $\gamma(0) = x_0$ and $\gamma(1) = x_1$. Given a vector bundle $E \to M$, or even just over a neighborhood of a given curve, and an element

$s(0) \in E_{x_0}$, we wish to *parallel transport* $s(0)$ along γ to find a smooth section $s(t)$ with $s(t) \in E_{\gamma(t)}$ over γ which satisfies for $t \in I$

$$\nabla_{\dot{\gamma}(t)} s(t) = 0.$$

To make sense of this we first note that the velocity filed $\dot{\gamma}$, which is only defined at points of γ, can be extended (non-uniquely) to a vector field X on a neighborhood of γ. Hence we have an initial value problem on that neighborhood: Determine a section s so that $\nabla_X(s)$ vanishes along γ. This is a first order ODE (system) which has a unique solution for the initial value $s(0)$. Thus for any $t \in I$ we have the unique *parallel transport* $s(t)$ which is a smooth section of E over γ satisfying the above condition.

If we cover a general smooth (or even piecewise smooth) curve γ by coordinate neighborhoods and solve the ODE in each case, the uniqueness of the solutions on the coordinate neighborhoods yields a well-defined *parallel transport operator* for γ; if the initial element $s(0) =: Y$, then one denotes this by

$$P_0^t(\gamma)(Y)\,.$$

This is a linear map from $E_{\gamma(0)}$ to $E_{\gamma(t)}$ which usually depends on γ.

The ODE, which is mentioned above, arises as follows. Let $s = s_j e_j$ be the expession of s in a given frame. If $X = x_i \partial_i$, then

$$\nabla_X s = s_j x_i \nabla_{\partial_j}(e_i) + x_i \partial_i(s_j) e_i = 0$$

is the ODE system , i.e., the coefficient of e_i vanishes for $i = 1, \ldots, n$.

One can check that parallel transport *defines* the connection in the following sense. If X is a vector field and s is a section, then in fact

$$\nabla_X(s)(x_0) = \lim_{h \to 0} \frac{\left(P_{\gamma(0)}^{\gamma(h)}\right)^{-1} (s(\gamma(h)) - s(\gamma(0))}{h}$$

where γ is the integral curve of X through $x_0 = \gamma(0)$. A hint for showing this is to express $s(t)$ in a *parallel frame* which is defined in a neighborhood of γ, i.e., the elements e_i are parallel in the above sense.

Bilinear forms, bundle metrics, Riemannian and pseudo-Riemannian metrics

First we consider a bilinear form $b \in \mathrm{Bilin}(V)$ on a vector space V, i.e., a map $b : V \times V \to \mathbb{R}$ which is linear in both arguments. If $(e_1, \ldots, e_n)$ is a basis of V with associated dual basis $(f_1, \ldots, f_n)$, then for every i, j we define a bilinear form $f_i \otimes f_j$ by

$$f_i \otimes f_j(e_k, e_\ell) = \delta_{ij}^{k\ell}\,.$$

Later we will discuss tensors in more detail, but for now this is just a notation. Clearly every $b \in \mathrm{Bilin}(V)$ is uniquely described as a linear combination $b = b_{ij} f_i \otimes f_j$, i.e., $\{f_i \otimes f_j : 1 \le i, j \le n\}$ is a basis of $\mathrm{Bilin}(V)$.

The bundle version of the above goes in the expected way: If $E \to M$ is a vector bundle whose smooth structure is given by frames $\boldsymbol{e} = (e_1, \dots e_n)$ with smooth frame change, then the dual frames $\boldsymbol{f} = (f_1, \dots, f_n)$ give the smooth structure on the dual bundle E^*. Since frame change of frames of the type $f_i \otimes f_j$ is also smooth, we have natural smooth vector bundle structure on the bundle $\mathrm{Bilin}(E) = \dot{\cup}\mathrm{Bilin}(E_x)$.

In geometry we most often encounter *symmetric bilinear forms*, i.e., those with the property $b(u, v) = b(v, u)$ for all u, v in the vector space at hand. Since this symmetry condition is linear, we therefore have the subbundle $\mathrm{Bilin}_{\mathrm{sym}}(E)$ of symmetric bilinear forms in the full bundle of bilinear forms $\mathrm{Bilin}(E)$. Sections of this bundle are those of interest here.

Recall that a symmetric bilinear form b on vector space has a *signature* (p, q, r), where p is the dimension of a maximal positive subspace, q is the dimension of a maximal negative subspace and r is the dimension of the degeneracy

$$V^{\perp_b} = \{u \in V : b(u, v) = 0 \ \forall v \in V\}.$$

We are interested here in non-degenerate (symmetric) forms, i.e., when $r = 0$. Note that the condition *non-degenerate* is open. The set of positive (or negative) definite forms is an open cone. These form a cone subbundle of the bundle $\mathrm{Bilin}_{\mathrm{sym}}(E)$ of the vector bundle of symmetric forms. A section g of $\mathrm{Bilin}_{\mathrm{sym}}(E)$ with image in this cone subbundle is called a (Riemannian) *bundle metric* on E. If $E = TM$ is the tangent bundle, then such a g is called a *Riemannian metric* on M. If we only require that $g(x)$ is non-degenerate at every $x \in M$ and g is not positive definite, g is called a *pseudo-Remannian metric* on M. In the latter case it is important to emphasize that while sums $\lambda_i g_i$ of Riemannian metrics are again positive definite, but in the pseudo-Riemannian case this is always the case. Thus, for example, it is easy to construct Riemannian metrics using partitions of unity, but the same technique does not provide global Lorenzian metrics.

Metric connections

Here g denotes a Riemannian metric on M. One says that a connection ∇ on TM is *compatible with the metric* if the for X, Y, Z arbitrary vector fields it follows that

$$X(g(Y, Z)) = g(\nabla_X Y, Z) + g(Y, \nabla_X Z).$$

Note that this is a sort of product rule for differentiation. However, on the LHS of this equation we see the vector field X differentiating three objects that are moving, i.e., the two vector fields Y and Z which are indeed differentiated on the RHS, and the metric. The usual product rule would involve derivatives of all three objects! Thus this condition can be interpreted as saying (to be made precise later) that $\nabla g = 0$, namely that the metric is constant (parallel) with respect to ∇. We will see that this means that parallel transport is an isometry along any curve!

A metric connection on TM is required to satisfy another, perhaps more mysterious, condition, i.e., that of being *torsion free*: for all vector fields X, Y and Z

$$\nabla_X Y - \nabla_Y X - [X, Y] = 0.$$

The LHS of this equation is often referred to the *torsion* and therefore the additional required condition is that of being *torsion free*. Such a connection is called a *metric connection* or the *Levi-Civita connection* associated to the metric g.

Theorem 1. *Associated to any Riemannian metric on M there is a unique Levi-Civita connection.*

The proof of this theorem is a bit painful. A sketch, some details of which were given in the lecture, goes as follows. First assume that ∇ is a Levi-Civita connection. Write down the quantity

$$\text{LHS} = Xg(Y,Z) + Yg(Z,X) - Zg(X,Y)$$

and expand it using the compatibility condition. Try to eliminate as ∇ from as many terms as possible using the vanishing of torsion. After doing this we have

$$2g(\nabla_X Y, Z) + \text{junk}$$

where the junk doesn't involve ∇. So we have *Koszul's formula*

$$g(\nabla_X Y, Z) = \frac{1}{2}\big(\text{LHS} - \text{junk}\big)$$

which proves the uniqueness. It also gives a formula for the ∇; so we must check that this gives us a connection, that it it is compatible with the metric and that it is torsion free. We lock ourselves in a closet and don't come out until we have done this. $\square$

Introduction to Differential Geometry

Professor Alan Huckleberry

Jacobs Univestiy Bremen

Semester: Spring 2019

Friday, 15. March 2019

Lectures: 7–9

Geodesics are discussed from the point of view of a connection as well as the length/energy functional. The Euler-Lagrange equations are derived. The Legendre transform is introduced and the Hamitonian equations on the cotangent bundle are derived. This is followed by a description of the canonical Liouville 1-form λ and the canonical sympletic 2-form ω_{std} on T^*M. The connection of the Hamiltonian equations of geodesic flow to the symplectic geometry of T^*M is indicated. For this we recommend reading the relevant sections of the books of Jost, Arnold and McDuff/Salamon.

Geodesics and parallel transport

Let $\gamma : [a, b] \to M$ be a smooth curve which for simplicity we assume to have image in a coordinate patch. Thus $\gamma(t) = (x^1(t), \ldots, x^n(t))$. One says that a vector field $V = V\big(\gamma(t)\big) = V^i(\gamma(t))\big)\delta_j$ is parallel along γ whenever $\nabla_{\dot\gamma(t)}V = 0$. In coordinates this means that

$$\big(\dot x^i V^j \Gamma^k_{ij} + \dot V^k\big) = 0\,.$$

Given the initial data $\big(\gamma(a), V(\gamma(a))\big)$ we regard this as an ODE for the parallel transport of $V\big(\gamma(a)\big)$ along γ.

A curve is a *geodesic* (with respect to the connection) if the parallel transport of $\dot\gamma(a)$ agrees with $\dot\gamma(t)$. The ODE for this curve is second order ODE

$$\dot x^i \dot x^j \Gamma^k_{ij} + \ddot x^k = 0\,.$$

This is interpreted in the usual way as first order ODE on the tangent bundle.

Length and energy functionals

Given x, y in a Riemannian manifold (M, g) (which is assumed to be connected) and $\gamma : [a, b] \to M$ a (piecewise) smooth curve from x to y, one defines

$$L(\gamma) = \int_a^b \|\gamma\|_g dt\,.$$

The distance $d(x, y)$ is defined to be the infimum over all such curves. It is an elementary fact, proved by a local comparison to the standard Euclidean distance, that d is a distance function on M which defines the given manifold topology. It

should also be noted that L is invariant by change of the parameter t, i.e., if $\varphi : [\alpha, \beta] \to [a, b]$ is a diffeomorphism, then $L(\gamma) = L(\gamma \circ \varphi)$. Thus we may assume that the curves under consideration are parameterized by arc length.

The energy functional is defined by

$$E(\gamma) = \int_a^b \|\dot{\gamma}(t)\|^2 dt\,.$$

It is important that if γ is parameterized by arc length, then $L(\gamma) = E(\gamma)$. Thus, assuming that L is minimized by a (piecewise) smooth curve (which we can then parameterize by arc length) we turn to studying the critical values of E.

Euler-Lagrange equations

We follow Arnold's notation:

$$\Phi(\gamma) = \int_{t_0}^{t_1} L\big(x(t), \dot{x}(t)\big) dt$$

where γ is a smooth curve in a coordinate chart which we regard as an open subset of $\mathbb{R}^n$. Following the usual first considerations in the *Calculus of Variations*, we consider all small perturbations $\gamma + h$ where $h(t_0) = h(t_1) = 0$, and, assuming L is smooth, write

$$\Delta(\gamma, h) := \Phi(\gamma + h) - \Phi(\gamma) = \int_{t_0}^{t_1} \Big(L(x + h, \dot{x} + \dot{h}) - L(x, \dot{x}) \Big)\, dt\,.$$

Using the Taylor develpment of L,

$$\Delta(\gamma, h) = \int_{t_0}^{t_1} \left(\frac{\partial L}{\partial x} h + \frac{\partial L}{\partial \dot{x}} \dot{h} + O(2) \right) dt\,.$$

Integrating by parts and using the fact that $h(t_0) = h(t_1) = 0$,

$$\Delta(x, h) = \int_{t_0}^{t_1} \left(\frac{\partial L}{\partial x} - \frac{d}{dt} \left(\frac{\partial L}{\partial \dot{x}} \right) \right) h + O(2)\,.$$

Let us say that γ is *extremal* if

$$\lim_{h \to 0} \frac{\Delta(x, h)}{\|h\|} = 0$$

for all h. One can show (using an argument with cutoff functions h) that γ is extremal if and only if the Euler-Lagrange equations

$$\frac{\partial L}{\partial x} - \frac{d}{dt} \left(\frac{\partial L}{\partial \dot{x}} \right) = 0$$

are fulfilled. At the level of the tangent bundle this is a first order ODE.

Two computations

One computes that the Euler-Lagrange equations are exactly the same as the equations for γ to be a geodesic with respect to the Levi-Civita connection. One also computes in the tangent bundle $\|\cdot\|^2$ is an invariant of the local group action of the Euler-Lagrange ODE, in particular if the initial velocity $\dot{\gamma}$ has unit length, then it remains unit length along the extremal curve and it follows that γ is extremal for the length functional as well.

Legendre Transform

The Legendre transform will allow us to transform the Euler-Langrange ODE on TM to a Hamiltonian ODE on T^*M. The later is not only simpler but, more importantly, can be formulated in terms of the canonical symplectic structure on T^*M. For more details in this matter we refer to Arnold. Remark: Arnold's notation can be interpreted for a vector valued variable x. Given our notation above, where x is a coordinate on M, using Arnold's notation of x is a bit confusing, because we will introduce a fiberwise Legendre transform on the tangent bundle where the (linear) coordinate is denoted by $\dot{q}$. Nevertheless we will stay with x in this paragraph.

Let V be an $\mathbb{R}$-vector space with coordinate x. A linear functional $p \in V^*$ on the dual space acts by $p(x) = px$. Let f be a convex function on V and consider the problem of maximizing the difference $px - f(x)$ on the submanifold defined by $y = f(x)$. Using the method of Lagrange multipliers one immediately observes that the value $x = x(p)$ which does the job, i.e., $px - f(x)$ is (uniquely) maximized at $x(p)$ is implicitly defined by $p = \frac{\partial f}{\partial x}(x(p))$. The Legendre transform is then given by the difference

$$\mathcal{L}(f)(p) = px - f(x)$$

where $x = x(p)$. We now apply this as a fiberwise map which associates the function $H = \mathcal{L}(L)$ on T^*M to a convex Langrangian L on TM. One refers to H as the associated Hamiltonian.

The associated Hamiltonian and its ODE

Here we carry out the above discussion for L a Lagrangian which defines the functional $\Phi(\gamma) = \int_{t_0}^{t_1} L(q, \dot{q})dt$ where in the usual way $(q, \dot{q})$ are coordinates in TM. It should, however, be emphasized that the Legendre transform $\mathcal{L}$ is globally defined. In T^*M we use the *momentum* coordinate p so that

$$H = H(p, q) = \mathcal{L}(L)(p, q) = p\dot{q} - L(\dot{q})$$

where $\dot{q} = \dot{q}(p)$ as above. Above we have the Euler-Lagrange equations and we will now derive the corresponding equations on T^*M. For this the first step is formal,

$$\frac{\partial H}{\partial p} = \dot{q},$$

and

$$\frac{\partial H}{\partial q} = -\frac{\partial L}{\partial q}\,.$$

Applying the Euler-Lagrange equation, the latter equation becomes

$$\frac{\partial H}{\partial q} = -\dot{p}\,.$$

Thus we obtain the Hamiltonian vector field

$$X_H = \frac{\partial H}{\partial p}\frac{\partial}{\partial q} - \frac{\partial H}{\partial q}\frac{\partial}{\partial p}$$

with associated ODE given by

$$\dot{q} = -\frac{\partial H}{\partial p} \text{ and } \dot{p} = \frac{\partial H}{\partial q}\,,$$

called the Hamiltonian ODE.

The contact and symplectic structures on T^*M

Let us first construct the canonical Liouville (contact) 1-form λ on T^*M. For this let $\pi : T^*M \to M$ be the canonical projection and τ be a tangent vector at a point $\alpha \in T^*_xM$. Then $\pi_*(\tau) \in T_xM$ and therefore $\tau \mapsto \alpha(\pi_*(\tau))$ defines $\lambda(\tau)$. The canonical (standard) 2-form is defined by $\omega_{std} = -d\lambda$. Let us compute these forms in coordinates.

For these computations we require the fiber coordinates $p = (p^1, \ldots, p^n)$ which will depend on the base coordinates by expressing a 1-form α in the fiber over a point with coordinates q as

$$\omega = \sum p_i dq^i \text{ so that } p(\omega) := (p^1(\omega), \ldots, p^n(\omega))\,.$$

It follows that $\lambda := pdq$ and $\omega_{std} := dq \wedge dp$. This is perhaps confusing notation and hence we emphasize hire that the dq^i in this notation are not 1-forms on M but rather the pull-backs of the corresponding 1-forms to T^*M!

The Hamiltonian equations determined by the symplectic structure

We will discuss this matter, in particular the basics of symplectic geometry, in the next lectures. For now we define the notion of a symplectic manifold and discuss the map $H \mapsto X_H$ from functions H to their Hamiltonian vector fields X_H.

A symplectic manifold is a pair (M, ω) where ω is a closed non-degenerate 2-form on the manifold M. Here non-degenerate means that at every point $m \in M$ the mapping $T_mM \to T^*_MM$, $v \mapsto \omega(v, \cdot)$, is an isomorphism. If (M, ω) is a symplectic manifold and $H \in C^\infty(M)$ is a function, then the associated vector field X_H is defined by

$$\iota_{X_H}\omega = \omega(X_H, \cdot) = dH\,.$$

The ODE associated to X_H is called the Hamiltonian equation defined by the Hamiltonian H. The connection to the above discussion is that if $H = \mathcal{L}(L)$ is the Legendre

transform of a convex Lagrangian and ω_{std} is the standard symplectic structure on the cotangent bundle T^*M, then X_H is the vector field

$$X_H = \frac{\partial H}{\partial p}\frac{\partial}{\partial q} - \frac{\partial H}{\partial q}\frac{\partial}{\partial p}\,.$$

which is the transformed Euler-Lagrange equation.

Introduction to Differential Geometry

Professor Alan Huckleberry

Jacobs Univestiy Bremen

Semester: Spring 2019

Friday, 22. March 2019

Lectures: 10–13

Review

Recall that a symplectic manifold is a pair (M, ω) where ω is a non-degenerate closed 2-form on M. Our first example, $(T^*M, \omega_{\text{std}})$, appeared in the previous lecture. Recall that $\omega_{\text{std}} = dq \wedge dp$ in canonical coordinates (q, p). For example, the Legendre transformed ODE for geodesics amounts to integrating the vector field

$$-\frac{\partial H}{\partial p}\frac{\partial}{\partial q} + \frac{\partial H}{\partial q}\frac{\partial}{\partial p}$$

on the cotangent bundle where in this case H is (up to a constant) the dual norm-squared. We now put this in the general framework of Hamiltonian *flows* on symplectic manifolds where by flow we mean the local group action which is associated to a vector field. We recall that if a vector field has compact support, e.g., if the ambient manifold is compact, then the associated local group action is a (global) $(\mathbb{R}, +)$-action.

Symplectic vector spaces

A pair (V, ω) of a vector space (in our case finite-dimensional) and a non-degenerate alternating bilinear form is called a *symplectic vector space*. As usual, for any basis and v, w in V represented as numerical column vectors in that basis, we may write $\omega(v, w) = v^t A w$ for an antisymmetric matrix A. As a concrete example, if W is any vector space, then $V := W \oplus W^*$ is equipped with a standard symplectic form ω_{std} which vanishes on W and W^* (they are Lagrangian subspaces) and $\omega_{\text{std}}(w, f) := f(w)$ for $w \in W$ and $f \in W^*$.

One good way of understanding a given a symplectic vector space (V, ω) is to note that, whereas diagonalization is impossible, we can decompose V as a direct sum $P_1 \oplus \cdots \oplus P_n$ of symplectic 2-planes where the restricted forms are non-degenerate. Note that (P, ω) is a symplectic plane, then it is immediate that it possesses a basis (e, f) with $\omega(f, e) = 1$. In this basis

$$\text{Mat}(\omega) = J = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}.$$

To find such a decomposition, let L_1 be any 1-dimensional subspace of V (a line) and W be its orthogonal complement $L_1^\perp$ with respect to ω. Since W is 1-codimensional,

there is a line L_2 in the complement of W. It follows that $P_1 = L_1 \oplus L_2$ is a symplectic plane. Since $\omega|P_1$ is non-degenerate, we have the decomposition $V = P_1 \oplus P_1^\perp$ and we continue by induction on dimension. Thus $\mathrm{Mat}(\omega) = J_1 \oplus \cdots \oplus J_n$ and in particular we see that $\dim(V) = 2n$. Of course this implies that symplectic manifolds are likewise even-dimensional.

No local invariants

It is seldom that two Riemannian manifolds (M_1, g_1) and (M_2, g_2) are locally isomorphic, but two symplectic manifolds of the same dimension are locally the same. The essential theorem in this direction, which is called Darboux's theorem (see the proof of J. Moser), can be stated as follows: Given a point x in a symplectic manifold (M, ω) there is a neighborhood $U = U(x)$ and a diffeomorphism $\varphi : U \to T^*M$ onto an open neighborhood of $0 \in T^*M$ with $\varphi^*(\omega_{\mathrm{std}}) = \omega|U$.

First examples

If (M, ω) is 2-dimensional then one can regard ω as an orientation form on M. Conversely, orientable 2-dimensional manifolds are symplectic. Note that for an arbitrary symplectic manifold (M, ω) of dimension $2n$ the n-fold wedge-product $\omega \wedge \cdots \wedge \omega$ is an orientation form on M. Hence, every symplectic manifold is orientable.

Since it is an orientable surface, the 2-sphere S^2 has a symplectic structure. This is the only example of a symplectic sphere. To see this, observe that if a $2n$-dimensional compact manifold M has a symplectic structure ω, then ω can not be exact, i.e., it is not of the form $\omega = d\mu$, because if it were, then the associated volume form would be

$$\Omega := \omega \wedge \cdots \wedge \omega = d(\mu \wedge \omega \wedge \cdots \wedge \omega)\,.$$

Thus by Stokes Theorem

$$0 \neq \int_M \Omega = \int_{\partial M} \mu \wedge \omega \wedge \cdots \wedge \omega\,,$$

and since $\partial M = \emptyset$, this integral must vanish. This can be formulated as follows.

Proposition 1. *If (M, ω) is symplectic and M is compact, then $[\omega] \neq 0 \in H^2_{deR}(M)$.*

Cartan's formula

Cartan's formula is an essential tool for the first steps of symplectic geometry. It states that the Lie derivative $\mathcal{L}_X : \mathcal{E}^k(M) \to \mathcal{E}^k(M)$ at the level of k-forms (any $k \geq 0$) is given by

$$\mathcal{L}_X = di_X + i_X d\,.$$

In other words $\mathcal{L}_X$ is equal to the *anti-commutator* of the differential d and contraction i_X.

Here is a sketch of the proof. Note that it is enough to give a proof in local coordinates for a form of the simple type $f dx_I = f dx_{i_1} \wedge \cdots \wedge dx_{i_k}$. This goes by

induction on k where the beginning of the induction follows immediately from the definition of d.

1. $(i_X d)(f dx_I) = X(f)dx_I - df \wedge i_X dx_I$
2. $(di_X)(f dx_I) = df \wedge i_X dx_I + f di_X(dx_I)$
3. RHS $= X(f)dx_I + f di_X(dx_I)$
4. LHS $= X(f)dx_I + f\mathcal{L}_X dx_I$
5. $\mathcal{L}_X dx_i = d\mathcal{L}_X \hat{x}_I$ where $\hat{x}_I = x_{i_1} dx_{i_2} \wedge \cdots \wedge dx_{i_k}$
6. Induction implies $d\mathcal{L}\hat{x}_I = di_X dx_I$
7. LHS = RHS is immediate

The basic exact sequence

Recall that a chain

$$A \xrightarrow{\alpha} B \xrightarrow{\beta} C$$

of morphisms is said to define a *complex* (at B) if $\mathrm{Im}(\alpha) \subseteq \mathrm{Ker}(\beta)$. One can say that the equations given by β vanish on the image of α, i.e., in order for an element of B to be in the image of α it must satisfy these equations. It is said to be *exact* (at B) if $\mathrm{Im}(\alpha) = \mathrm{Ker}(\beta)$, i.e., if the equations given by β exactly describe the image of α.

The first important step for the basic sequence is map h from the smooth functions on M to vector fields of a special type: $h(f) := X_f$ where $df = \omega(X_f, \cdot) = i_X(\omega)$. The *special type* property of the vector field X_f is given by

$$\mathcal{L}_{X_f}(\omega) = di_{X_f}\omega + i_{X_f} d\omega = 0$$

because $d\omega = 0$ (by the definition of symplectic manifold) and $i_{X_f}\omega = df$ (by the definition of X_f). This means that that the local 1-parameter group (the flow) of X_f preserves ω. Since a 2-form can be regarded as a measure of area, it is in this sense that one says that the *Hamiltonian flow preserves area*. We let $\mathrm{Ham}_{\mathrm{loc}}(M)$ denote the space of vector fields X which have this property, i.e., $\mathcal{L}_X(\omega) = 0$. Thus, the mapping h can be regarded as a mapping $h : C^\infty(M) \to \mathrm{Ham}_{\mathrm{loc}}(M)$. Its image is denoted by $\mathrm{Ham}(M)$, the space of Hamiltonian vector fileds.

Let us try to pin down the image of h in $\mathrm{Ham}_{\mathrm{loc}}(M)$. For this note that if $X \in \mathrm{Ham}_{\mathrm{loc}}(M)$, another application of Cartan's formula shows that $\mu_X = i_X(\omega)$ is a d-closed 1-form. So we introduce the de Rham map $\mathrm{deR} : \mathrm{Ham}_{\mathrm{loc}}(M) \to H^1_{\mathrm{deR}}(M)$, $X \mapsto [i_X\omega]$. In fact this map is surjective and consequently we have the sequence

$$0 \longrightarrow \mathbb{R} \xrightarrow{i} C^\infty(M) \xrightarrow{h} \mathrm{Ham}_{\mathrm{loc}}(M) \xrightarrow{\mathrm{deR}} H^1_{\mathrm{deR}}(M) \longrightarrow 0\,.$$

To review, i is the canonical injection which maps a number to the constant function of that value (here we have assumed that M is connected in order to prove exactness at that level), $h(f) = X_f$ and deR associates X to the de Rham class of $i_X\omega$.

Constants of motion

A smooth function f is said to be a constant of motion of a vector field X (or its associated local 1-parameter group) whenever $X(f) = 0$. It is a nice exercise to show that this is the same thing as requiring that the orbits of the local action, i.e., the integral curves, stay inside the level sets $\{f = c\}$ for all values c. For any Hamiltonian field X_H it is immediate that $X_H(H) = 0$, because

$$X_H(X) = dH(X_H) = \omega(X_H, X_H) = 0\,.$$

For example, it is often the case that H is the total energy and in such a case one refers to this as *Conservation of Energy*. In the case of geodesic flow on the cotangent bundle H is (up to constants) the dual norm-squared $\|\cdot\|^2_{g^*}$ and therefore this norm is preserved by geodesic flow. In concrete terms, the total space Σ of the unit sphere bundle in the cotangent T^*M is preserved by geodesic flow. Note that if M is compact, then so is Σ and consequently geodesic flow is a global $(\mathbb{R}, +)$ in that case. A basic theorem (Hopf-Rinow) states that if M is complete with respect to the distance function coming from the Riemannian metric, then, even though M might not be compact, geodesic flow is global in this sense.

The associated Poisson structure

It is prudent to think of the vector field X_f as defining an operator $\{f, \cdot\}$ on the smooth functions which is defined by $\{f, g\} = -X_f(g)$. It follows that

$$\{f, g\} = \omega(X_g, X_f)\,.$$

Since ω is alternating, it is immediate that $\{\cdot, \cdot\}$ is anti-symmetric and the fact that H is a constant of motion of X_H is simply the fact that $\{H, H\} = 0$.

The fact that $d\omega = 0$ is equivalent to the Jacobi identity for this *Poisson bracket*, namely that $\{f, \cdot\}$ is a derivation in the sense that

$$\big\{f, \{g, h\}\big\} = \big\{\{f, g\}, h\big\} + \big\{g, \{f, h\}\big\}\,.$$

In other words, $(C^\infty(M), \{\cdot, \cdot\})$ is a Lie algebra. The space of functions is of course not just a vector space, because two functions can be multiplied: $(f, g) \mapsto fg$ is a commutative multiplication. It is not surprising that $\{f, \cdot\}$ behaves as a first order differential operator: $\{f, gh\} = \{f, g\}h + g\{f, h\}$. A Lie algebra with this kind of structure is referred to as a *Poisson algebra*.

The full space of constants of motion of a Hamiltonian flow X_H is simply the centralizer of H in the Poisson algebra:

$$Z(H) = \big\{f \in C^\infty(M) : \{H, f\} = 0\big\}\,.$$

The elements $f \in Z(H)$ are those functions which commute with H. Of course, the bigger the centralizer the better. For example, if $H = H_1$ and H_2 commute and are analytically independent in the sense that $dH_1 \wedge dH_2 \neq 0$, then at a point x where $dH_1 \wedge dH_2(x) \neq 0$ both of the level surfaces $M_i = \{H_i = H_i(x)\}$, $i = 1, 2$, are smooth and intersect transversally in a smooth submanifold $M_1 \cap M_2$ which is invariant under the both flows of X_{H_i}.

Arnold-Liouville and KAM

Of course the existence of high-codimensional stable manifolds is a weaker requirement than that of a large number of independent constants of motions, but nevertheless considering the latter possibility is good first start at understanding *something*. From this point of view the absolute optimal situation for a $2n$-dimensional symplectic manifold is that of *complete integrability* where we have n Hamiltonians $H_1, \ldots, H_n$ in involution, i.e., $\{H_i, H_j\} = 0$ for all i and j. One can regard a given Hamiltonian H to be H_1 and all of the H_i be constants of motion of H, but due to the symmetry of the condition (all being in involution), this is irrelevant. If the H_i are analytically independent and define as joint level set $N = \{H_1 = c_1, \cdots, H_n = c_n\}$, then the associated vector fields $X_i = X_{H_i}$ are tangent to N. Since they commute, it is easily shown that if N is compact, it is isomorphic to a torus with the integrated flows of the X_i begin linear. Since the X_i commute, it is immediate that the restriction of ω to N vanishes identically, i.e., N is a Lagrangian submanifold of M.

Of course one would hope that the H_i would define a family of Lagrangian tori. Arnold formulated this in an optimal way in terms of a Lagrangian fibration and then proved that such a fibration is just a (fiberwise) quotient of the cotangent bundle T^*Q of a base (configuration space) Q which parameterizes the fibers. This quotient is given fiberwise by a lattice Γ_q which in general is varying with q. This classification theorem, which is called the Theorem of Arnold-Liouville, shows that in a certain sense we have n independent harmonic oscillators and therefore the (q, p) coordinates coming from the cotangent bundle are often called *angle-action* coordinates.

The above theorem describes an idealized situation which would be difficult to find in nature. Therefore, it is of great importance to understand if it is in *stable* under small perturbations of the ingredients. The Theorem of KAM (Kolmogorov-Arnold-Moser) gives a certain answer to this: under a precisely defined type of small perturbation certain tori survive and are stabilized by the Hamiltonian system at hand.

Hermitian structures

If we start with the standard symplectic vector space $V = W \oplus W^*$ with $(e_1, \ldots, e_n)$ a basis of W, $(f_1, \ldots, f_n)$ the dual basis and $\omega(f_i, e_j) = f_i(e_j) = \delta^i_j$ as above, then for numerical vectors v, w in the basis $(e_1, \ldots, e_n, f_1, \ldots, f_n)$ of V the symplectic form is given by

$$\omega(v, w) = v^t J w$$

with

$$J = \begin{pmatrix} 0 & Id \\ -Id & 0 \end{pmatrix}.$$

If $g(v, w) := \omega(Jv, w)$, then g is a symmetric positive definite bilinear form. In fact in this basis its matrix is just the identity. Finally,

$$h(v, w) := g(v, w) - i\omega(v, w)$$

is seen to be a (non-degenerate) Hermitian form on V: $h(w, v) = \overline{h(v, w)}$ and $h(Jv, w) = ih(v, w)$.

The above is an example of the following general principle. If (V, J) is a $\mathbb{R}$-vector space with a complex structure J, i.e., $J^2 = -Id$ and ω is a linear symplectic form on V, then one says that J is ω-tame if $\omega(Jv, Jw) = \omega(v, w)$ for all v and w and the associated symmetric form $\omega(Jv, w) =: g(v, w)$ is positive definite. It follows that the form $h = g - \omega$ is Hermitian and positive definite. In fact, we could have started with a Riemannian (symmetric) form and made an analogous definition to obtain ω.

Let us discuss the analogous definition at the level of manifolds. One can first define the notion of a complex structure J on the tangent bundle TM. Such is just a bundle endomorphism $J : TM \to TM$ which satisfies $J^2 = -Id$ where here we are requiring it to be linear in the fibers but trivial on the base. This is a sort of infinitesimal complex structure. If M is a complex manifold in the sense that it has a covering of coordinate patches equipped with compatible holomorphic coordinates, then we also have an complex structure J: If $(z^1, \ldots, z^n)$ are holomorphic coordinates with $z^j = x^j + iy^j$, then we can define J at the frame level in the expected way by

$$J\left(\frac{\partial}{\partial x^j}\right) = \frac{\partial}{\partial y^j} \quad \text{and} \quad J\left(\frac{\partial}{\partial y^j}\right) = -\frac{\partial}{\partial x^j}\,.$$

Holomorphic changes of coordinates φ then satisfy $\varphi_* \circ J = J \circ \varphi_*$ which is the statement that φ_* is complex linear with respect to J. A manifold with a complex structure J in this infinitesimal sense has holomorphic coordinates if and only if the Nijenhuis tensor vanishes for every pair of vector fields X and Y:

$$N(X, Y) = [JX, JY] - J[JX, Y] - J[X, JY] - [X, Y] = 0\,.$$

Given an infinitesimal complex structure J which satisfies this condition one must solve an associated pde to find the holomorphic coordinates. This is the Theorem of Newlander-Nirenberg.

If (M, ω) is a symplectic manifold, then an infinitesimal complex structure J is said to be ω-tame if $\omega(JX, JY) = \omega(X, Y)$ and $g(X, Y) := \omega(JX, Y)$ is a positive definite Riemannian metric. Thus, starting with a symplectic structure, we obtain a Riemannian and Hermitian structure $h = g - i\omega$. The full space of all ω-tame structures is an infinite dimensional (contractible) manifold. In particular, one has all sorts of curves J_t of such structures. On the other hand, the space of complex manifold structures should be regarded as being finite-dimensional, maybe even empty! One has much more freedom when dealing with ω-tame structure and furthermore they provide important associated Riemannian and Hermitian geometries.

Kähler manifolds

In the 1930's E. Kähler (Hamburg) called attention to manifolds M of the above type where one assumes that the complex structure J comes from coordinates and is ω-tame for a symplectic structure ω. One can show that locally these Kähler forms can be given by $\omega = (i/2)\partial\bar{\partial}\rho$ where ρ satisfies additional conditions in order for g (as above) to be positive definite. It had been know for many years that these *strictly plurisubharmonic* functions are of great importance in complex analysis. Nowadays one refers to such a manifold (M, ω) as a Kähler manifold.

Since $h = g - i\omega$, one can view Kähler manifolds as being complex manifolds which possess special Hermitian (or Riemannian) metrics. It is a simple matter to check that if complex manifold X is Kählerian with respect to ω, then every complex submanifold Y is Kählerian with respect to the restricted form $\omega|Y$. In the next lecture we will introduce the Fubini-Study Kähler metric ω_{FS} on any given complex projective space $\mathbb{P}_n(\mathbb{C})$. Thus an important class of Kähler manifolds consists of those which can be realized as complex submanifolds of projective space. This mix of Kählerian (differential) and algebraic geometry has led to our understanding of numerous phenomena, some of which will be discussed later on in the course.

Pseudoholomorphic curves

Gromov (see his basic 1985 paper in Inv. Math.) turned the above idea around by emphasizing the importance of a complex geometric viewpoint in symplectic geometry. We will not go into this deep and very important subject at any level of seriousness, but do want to underline one concept, i.e., that of a *pseudoholomorphic curve*.

We start with a symplectic manifold (M, ω) where, for example, one wishes to understand (or even simply define) invariants of Hamiltonian flows. Here we have a huge space of ω-tame complex structures J. Thus we have a degree of flexibility which is not available in complex analysis, e.g., one easily builds curves J_t of such structures. At first let us fix such a structure so that we have a triple (M, J_M, ω) and let us fix a compact Riemann surface S (complex structure J_S with coordinates). Then we have the notion of a holomorphic map $\varphi : S \to M$: $\varphi_*(J_S v) = J_M \varphi_*(v)$ at the level of tangent bundles. We regard this as a curve (analogous to a geodesic) in M. Since the complex structure on the target manifold is only infinitesimal, such curves are called *pseudoholomorphic*. Spaces of pseudoholomorphic curves, which are related to the Riemannian viewpoint by $h = g - i\omega$, are of fundamental importance for many modern developments in symplectic geometry.

Introduction to Differential Geometry

Professor Alan Huckleberry

Jacobs Univestiy Bremen

Semester: Spring 2019

Friday, 29. March 2019

Lectures: 14–17

Elementary computations with Kähler forms

Recall that a Kähler manifold is a complex manifold M, i.e., the complex structure J is defined by holomorphic coordinates, with a symplectic form ω, called the Kähler form, such that J is $\omega-$tame. In order to get the signs right we should look at the case where M is just $\mathbb{R}^2$ equipped with

$$\omega = dx \wedge dy = \frac{i}{2} dz \wedge d\bar{z}\,.$$

The complex structure is given by J with

$$J(\frac{\partial}{\partial x}) = \frac{\partial}{\partial y} \quad \text{and} \quad J(\frac{\partial}{\partial y}) = -\frac{\partial}{\partial x}\,.$$

We observe, correcting a sign mistake in the last lecture, that the associated Riemannian metric should be defined by

$$g(v, w) = \omega(v, Jw)$$

and with this convention g is a (positive-definite) Riemannian metric with $(\frac{\partial}{\partial x}, \frac{\partial}{\partial y})$ an orthonormal frame.

Recall also that the Kähler condition requires that $\omega(Jv, Jw) = \omega(v, w)$. In this 2-dimensional setting this is automatically fulfilled, because the space of 2-forms is spanned (over the smooth functions) by the above form. In higher dimensions this is not the case, e.g., on $\mathbb{C}^2$ we have the form $dz_1 \wedge dz_2$. We may think of J as acting on forms in the obvious way with $J(dz_i) = idz_i$ and $Jd\bar{z}_i = -idz_i$. If a form can be written as

$$\omega = \sum_{|I|=p, |J|=q} a_{IJ} dz^I \wedge d\bar{z}^J$$

in multi-index notation, then we refer to it as a $(p, q)-$form. This notion is canonical and has the consequence that the above condition on a Kähler form amounts to ω begin a $(1, 1)-$form. Of course it must be real and the associated real bilinear form g must be positive definite.

Kodaira's Lemma summarizes the above in a nutshell: Every Kähler form is locally of the form

$$\omega = \frac{i}{2} \partial\bar{\partial}\rho$$

where ρ is a real valued function. Of course not every real valued function ρ defines a Kähler form in this way, because the further condition that g is positive definite in general not satisfied. If it is satisfied, ρ is called *strictly plurisubharmonic*. Such functions play a major role in many aspects of complex geometry.

Lie groups and their actions

The fiber bundle $\mathbb{C}^{n+1} \setminus \{0\} \to \mathbb{P}_n(\mathbb{C})$, which arises when defining the complex projective space $\mathbb{P}_n(\mathbb{C})$, is of basic importance in Kähler (algebraic) geometry. It is defined by the $\mathbb{C}^*$-action defined by $\lambda(v) = \lambda v$ on $\mathbb{C}^{n+1}\{0\}$ with the bundle being the quotient by this action. We will push down a certain $(1,1)$-form defined on the *upstairs space* $\mathbb{C}^{n+1} \setminus \{0\}$ to obtain an optimal Kähler form, the Fubini-Study form ω_{FS}, on the *downstairs* space $\mathbb{P}_n$. It is useful to think about this in the general context of principal bundles which requires several introductory remarks on Lie groups .

A Lie group is a paracompact differentiable (a fortiori real analytic) manifold which is a group G equipped with a smooth multiplication $G \times G \to G$, $(g,h) \mapsto gh$, and inverse $G \to G$, $g \mapsto g^{-1}$, maps. A left G-action on a manifold is a smooth map $\alpha : G \to M \to M$, $(g,x) \mapsto \alpha(g,x)$, so that the map $G \to \mathrm{Diff}(M)$, defined by $g \mapsto \alpha(g,\cdot) =: \alpha(g)$ is a homomorphism. In particular, $\alpha(gh) = \alpha(g)\alpha(h)$. If this homomorphism rule is the other way around, i.e., $\alpha(gh) = \alpha(h)\alpha(g)$, then one refers to this as a right action. We have already encountered such action: If $\boldsymbol{e}$ is a frame for a vector bundle then frame change is given by $\boldsymbol{e} \mapsto \boldsymbol{e}g$.

Various Lie groups appear in the very first steps of differential geometry. For example, if $\boldsymbol{e}$ is a frame of a vector bundle, then frame change $\boldsymbol{e} \mapsto \boldsymbol{e}g$ is given by a smooth mapping g into the Lie group $GL_n(\mathbb{R})$. If a vector bundle E on M is equipped with a metric b and $p \in M$, then $b(p)$ is a positive definite bilinear form on E_p and the group of linear isometries of E_p (which leave $b(p)$ invariant) is a compact Lie group which can be denoted by $O(E_p, b(p))$. It is called the orthogonal group of E_p with respect to $b(p)$.

Unitary groups are of particular importance in geometry and physics. Analogous to the orthogonal group defined above, if h is a positive definite Hermitian structure on a complex vector space V, then $U(V,h)$ is the group of complex linear transformations B of V which preserve h in the sense that $h(Bv, Bw)$ for all $v, w \in V$. Using the standard coordinates on $\mathbb{C}^n$, a Hermitian structure structure is defined by $h(z,w) = \bar{z}^t w$ and therefore B is a unitary matrix if $\bar{B}^t B = Id$. The group of such matrices is denoted by U_n and the subgroup of operators having determinant 1 is denoted by SU_n.

Quite often a mixed signature Hermitian structure is of importance. For example, if the norm is given by

$$\|z\|^2 = \sum_{i=1}^{p} |z_j|^2 - \sum_{i=1}^{q} |z_{p+j}|^2$$

with $p + q = n$, then the group of operators preserving this norm (or equivalently the associated Hermitian form) is denoted by $SU(p,q)$ (providing unit determinant is required).

One point with unit determinant becomes clear when we consider actions of Lie groups on projective space. For example $G = GL_{n+1}(\mathbb{C})$ acts on $\mathbb{P}_n(\mathbb{C})$ in the obvious way, but the subgroup of diagonal matrices $N := \mathbb{C}^* Id$ acts trivially. This is a normal (in fact central) subgroup and therefore we may reduce to an action of G/N. This is almost the same thing as reducing to the action of $SL_{n+1}(\mathbb{C})$, but not quite: for example $-Id$ is still in SL_2.

Of particular importance for us are the groups SU_2, and its action on $\mathbb{P}_1(\mathbb{C}) \cong S^2$, and $SU(1,1)$, and its action on the unit disk Δ. In the former case this action is defined by the SU_2 action on $\mathbb{C}^2$ where it acts freely and transitively on the standard spheres S^3. Thus, as a manifold, SU_2 is the 3-sphere; in particular it is compact. As usual the matrices of such a group can be described by an orthonormal (in this case we say *unitary*) basis of column vectors. For SU_2 we have

$$SU_2 = \begin{pmatrix} a & -\bar{b} \\ b & \bar{a} \end{pmatrix} \qquad \text{with } |a|^2 + |b|^2 = 1\,.$$

Observe that if we drop the norm-condition, the resulting set of such matrices is just $\mathbb{C}^2$. Equipped with its $\mathbb{R}$-linear operations and matrix multiplication, this is just space of quaternions.

The matrices B in $SU(1,1)$ satisfy $\bar{B}^t E_{1,1} B = E_{1,1}$ where $E_{1,1} = Diag(1,-1)$. They are of the form

$$B = \begin{pmatrix} a & \bar{b} \\ b & \bar{a} \end{pmatrix}.$$

The determinant condition in this case is $|a|^2 - |b|^2 = 1$ and we immediately see that this Lie group is not compact.

Principal bundles

Here, the example to keep in mind is the $\mathbb{C}^*$-prinicipal bundle $\mathbb{C}^{n+1} \setminus \{0\} \to \mathbb{P}_n(\mathbb{C})$. In general the total (upstairs) space P of a G-principal bundle is a manifold equipped with smooth G-action which is *proper* and *free*. The properness of the action implies in particular that the G-orbits in P are closed and that the quotient P/G is Hausdorff in its quotient topology. Since the action is in addition free, at every point $p \in P$ there is a locally closed submanifold S of complementary dimension to the G-orbit so that the restriction of the action mapping, $G \times S \to P$, defines a diffeomorphism onto an open neighborhood of p. This implies in particular that S can be identified with an open neighborhood of the corresponding point in the orbit space P/G. Regarding any such S as defining a coordinate chart on P/G one checks that this then defines a canonically associated differentiable structure on $M := P/G$. We let $\pi : P \to M$ denote the projection map which is then differentiable and refer to $\pi : P \to M$ as a principal G-bundle.

Pushing down differential forms defined on P

It is quite often the case that the total space P is naturally equipped with objects which have deep meaning on the base M. One simple object of this kind is a

smooth function f which is G-invariant, i.e., $g^*(f) = f$ for all $g \in G$. One checks that the image of the pull-back mapping $\pi^* : C^\infty(M) \to C^\infty(P)$ is exactly the space $C^\infty(M)^G$ of G-invariant functions. In other words, a function can be pushed down if and only if it is invariant.

For differential forms μ on P it is clear that G-invariance is a necessary condition for pushing down. However, one immediately notes that if $\mu = \pi^*(\omega)$, then the contraction $i_v(\mu)$ with a vector $v \in T_pP$ which is tangent to the fiber vanishes. Thus this second condition must be imposed. Let's prove the expected result.

Let μ be a k-form on P which is both G-invariant and has the above *contraction property*. We wish to define ω on M with $\pi^*(\omega) = \mu$. For this let $w = (w_1, \dots w_k)$ be a given k-vector at x and note that, since $\pi : P \to M$ is surjective and everywhere of maximal rank, there exists $v = (v_1, \dots, v_k)$ at any given $p \in P$ with $\pi(p) = x$ with the $v_i \in T_pP$ and $\pi_*(v) = w$. The point now is that there are other k-vectors at p, say $\hat{v}$, with the same property. But $\kappa_j := v_j - \hat{v}_j$ is in the kernel of π_*, i.e., κ_j is tangent to the π-fiber at p and the contraction property implies that ω defined by $\omega(w_1, \dots, w_k) := \mu(v_1, \dots, v_k)$ is well-defined independent of the choice of v. The invariance of μ shows that this discussion does not depend on the point $p \in \pi^{-1}(x)$. To see that ω is smooth we take a local section S and take v to be the unique choice of a lift of w which is tangent to S. One can check the converse of this statement to prove the following fact.

Proposition 1. *The image of the pullback map π^* at the level of k-forms is the space of k-forms on P which are G-invariant and satisfy the contraction condition.*

The Fubini-Study Kähler form on projective space

We attempt to find a $(1,1)$-form μ on $\mathbb{C}^{n+1} \setminus \{0\}$ which can be pushed down to a Kähler form on $\mathbb{P}_n(\mathbb{C})$. Motivated by Kodaira's Lemma and proceeding optimistically, we hope to find a globally defined function ρ so that $\mu = \frac{i}{2}\partial\bar{\partial}\rho$ does the job. Of course we can't hope that ρ will be strictly plurisubharmonic, e.g., the norm function $\|\cdot\|^2$, because μ must have the contraction property and in particular must vanish on the fibers of the principal bundle. It is interesting that $\rho = \ln(\|\cdot\|^2)$ not only yields μ having having the contraction property but also the invariance of μ: the multiplicative transformation rule $\|\lambda z\|^2 = |\lambda|^2\|z\|^2$ is turned into an additive rule by the logarithm and this is then annihilated by $\partial\bar{\partial}$.

The pushdown of μ is referred to as the Fubini-Study form ω_{FS} on $\mathbb{P}_n(\mathbb{C})$. In order to show that it is a Kähler form one much check that it is positive definite. As a first idea for this we note that for $n = 1$ in the coordinate $z = \frac{z_1}{z_2}$ a direct computation shows that

$$\omega_{\mathrm{FS}} = \frac{i}{2}\frac{dz \wedge d\bar{z}}{(1+|z|^2)^2}\,.$$

It should be remarked that up to irrelevant modifications ω_{FS} is the unique Kähler metric on $\mathbb{P}_n$ which is $SU_{n+1}-$invariant.

Now, let (X, ω) be a Kähler manifold and Y be a complex submanifold of X. We think of the tangent bundle TY as a subbundle of the restriction $TX|Y$. Since Y is locally defined by as $\{z_1 = \cdots = z_k = 0\}$, where $(z_1, \dots, z_n)$ are holomorphic coordinates of X, it is immediate that TY is J-invariant. Thus it is immediate

that the restriction (pullback by the inclusion) of ω to Y is a Kähler form on Y. In particular, every complex submanifold of $\mathbb{P}_n(\mathbb{C})$ comes equipped with a Kähler structure. In this way Kählerian differential geometry plays an important role in complex algebraic geometry.

A suggestion of Cartan

It is clear that on any given manifold M there is a huge space of Riemannian metrics and even more connections. So an interesting question is "On a given manifold, what would be an optimal metric?" For example, on S^2 we just discovered a unique SU_2-invariant metric! "In general, what would be a reasonable invariance condition?" Cartan's idea, which we will discuss in detail later, is to require that curvature is invariant by parallel transport. To dig into this, we will now introduce two ways of thinking about curvature, and in the next block of lectures explain the ramifications of Cartan's condition.

First remarks on curvature

If M is equipped with a connection ∇, then the associated Riemannian curvature tensor associated to two vector fields X and Y is defined as

$$R(X,Y) = \nabla_X \nabla_Y - \nabla Y \nabla X - \nabla_{[X,Y]}\,.$$

While this makes sense for any vector bundle, in the case of the tangent bundle this is a $(1,3)$-tensor as a map $W \mapsto R(X,Y)(W)$ of vector fields. This can be motivated by discussing parallel transport around the sides of a parallelogram defined by X and Y.

Another, perhaps technically simpler way of introducing essentially the same thing, goes as follows (for a general vector bundle E). Regard the connection as a map $D : \Gamma(M,E) \to \Gamma(M, E \otimes T^*M)$ where in this notation $\nabla_X(s) = i_X D(s)$. Given a frame $\boldsymbol{e}$ of E this (as usual) is described by a matrix $\omega(\boldsymbol{e})$ of $1-$forms: $De_\beta = e_\alpha \otimes \omega_{\alpha\beta}$. The change of frames by an element $g \in Gl_n(\mathbb{R})$ is not by conjugation as one might hope: If a new frame is given by $\boldsymbol{e}g$, then

$$\omega(\boldsymbol{e}g) = g^{-1}dg + g^{-1}\omega(\boldsymbol{e})g\,.$$

Now if d is the trivial connection on the trivial bundle, then we have a simple method of extending to the full exterior algebra with $d^2 = 0$. In the general case the only real choice of extension is given by $D(s \otimes \omega) := D(s) \wedge \omega + s \otimes d\omega$ as a map $D : \Gamma(E, E \otimes T^*M) \to \Gamma(E, E \otimes \wedge^2 T^*M)$. The curvature of D is then just the mapping D^2, i.e.,the obstruction to having a deRham-type complex.

We postpone the discussion the relation D^2 to the Riemannian curvature tensor. But we do note that for a frame $\boldsymbol{e}$ this D^2 is given by a matrix $F = d\omega + \omega \wedge \omega$. One refers to $F = F(e)$ as the associated connection 2-form. It is tensorial in the sense that $F(e.g) = g^{-1}F(e)g$.

Invariants of conjugation

In the above context it is appropriate to apply conjugation invariants to a curvature form to obtain objects with are frame-invariant. The most obvious invariant is perhaps *trace*: $Tr(F)$ is a frame invariant quantity. Another such invariant is the determinant $\det(F)$. In general the space of invariants of linear endomorphisms A form a polynomial algebra with generators $c_k = Tr(\wedge^k(A))$. In particular, in the case of the curvature form one immediately has the invariant "polynomials" $Tr(\wedge^k F)$.

Introduction to Differential Geometry

Professor Alan Huckleberry

Jacobs Univestiy Bremen

Semester: Spring 2019

Friday, 05. April 2019

Lectures: 18–21

Remarks on splittings

In preparation for a new view of a connection on a vector bundle, we review some aspects of the notion of *splitting*. Perhaps the most simple starting point for this is a subspace V of a (finite-dimensional) vector space W which defines a short exact squence

$$0 \longrightarrow V \xrightarrow{\alpha} W \xrightarrow{\beta} Q \longrightarrow 0$$

where $Q = V/W$ is the associated quotient space. The *splitting* question is whether or not there is a (linear) map $s : Q \to V$ with $\beta \circ s = Id$. In this case its image H is such that $W = V \oplus H$. Note that for vector space there are many such splittings. One way to create a splitting in this case is to introduce a positive-definite bilinear form b on W, define $H := V^{\perp_b}$, note that $\beta|H \longrightarrow Q$ is an isomorphism and then define $s : Q \to H \hookrightarrow W$ as its inverse.

For other algebraic structures the analogous questions are not so simple to answer. For example, if G is a group and N is a normal subgroup, then we have an associated exact sequence

$$0 \longrightarrow N \longrightarrow G \longrightarrow Q \longrightarrow 0$$

of groups. This may split and in that case there may be a number of interesting possibilities for splitting it. For example if $G = V_4$ and N is some subgroup isomorphic to $\mathbb{Z}_2$, then this is indeed the case. On the other hand if $G = \mathbb{Z}_4$ and N is the unique copy of $\mathbb{Z}_2$ embedded in G, then it does not split.

Another setting where the splitting is possible is that of vector bundles: Every short exact sequence

$$0 \longrightarrow E \longrightarrow F \longrightarrow Q \longrightarrow 0$$

(non-uniquely) splits and one way to construct such a splitting is by using a bundle metric on F.

A connection on a vector bundle as a splitting

If $E \xrightarrow{\pi} M$ is a vector bundle over a manifold M, then TE contains a (canonical) subbundle $\mathcal{V}$ which is the kernel of the surjective map $\pi_* : TE \to TM$. Since the tangent space of a vector space is the vector space itself and $\mathcal{V}_{v_x}$ is therefore canonically isomorphic to E_x, it follows that $\mathcal{V} = \pi^*(E)$.

A bundle splitting of

$$0 \longrightarrow \mathcal{V} \longrightarrow TE \longrightarrow Q \longrightarrow 0$$

has image $\mathcal{H}$ which is called a horizontal bundle. Since $TE = \mathcal{V} \oplus \mathcal{H}$, it defines (or in fact is equivalent to) a projection $\rho : TE \to \mathcal{V}$.

Now if $\sigma : M \to E$ is a section, then $\sigma_* : T_x M \to TE_{\sigma(x)}$ is an isomorphism onto its image. We follow this by ρ which, since $\mathcal{V} = \pi^* E$, defines a map

$$\rho \circ s_* : T_x M \to E_x \, .$$

Globallizing, if X is a vector field on M, then $\rho \circ s_*(X)$ is again a section of E and, using the classical notation, one defines this to be $\nabla_X s$.

In order to insure that this is a connection in our previous sense, we must impose a condition on the horizontal bundle $\mathcal{H}$, namely that it is compatible with the vector space structure of the fibers of E. For this we first note that scalar multiplication $S : \mathbb{R} \times E \to E$, $\lambda(v) := \lambda v$ is well-defined and therefore require that S_* stabilizes $\mathcal{H}$. Compatibility with addition goes analogously: Addition defines a bundle map $A : E \times_M E \to E$. The bundle $\mathcal{H}$ defines a horizontal bundle $\hat{\mathcal{H}}$ on the domain space of A and one requires that A_* maps $\hat{\mathcal{H}}$ to $\mathcal{H}$.

To emphasize the compatibility condition, let's refer to a horizontal bundle $\mathcal{H}$ which satisfies it to be *invariant* and the resulting connection to be an *invariant Ehresmann connection*. One checks that such a connection is exactly a connection in our previous sense (∇ or D) and conversely every connection in our previous sense defines an invariant Ehresmann connection.

A splitting defined by a submanifold

Let M be a submanifold of a manifold $\tilde{M}$ (immersed is enough) and consider the resulting exact sequence

$$0 \longrightarrow TM \longrightarrow T\tilde{M}|M \longrightarrow N_{\tilde{M}}(M) \longrightarrow 0$$

of vector bundles. The quotient bundle $N_{\tilde{M}}(M)$ is called the *normal bundle* of M in $\tilde{M}$. As we know, this sequence splits and one way to split it is with a bundle metric $\tilde{g}$ on $\tilde{M}$. Here we always assume that $\tilde{M}$ is equipped with such a Riemannian metric and let g denote the induced metric on M (we are following the notation in the textbook of Lee). Let $\tilde{\nabla}$ and ∇ denote the corresponding Levi-Civita connections. Using the splitting defined by $\tilde{g}$ we write

$$T\tilde{M}|M = TM \oplus_{\tilde{g}} N_{\tilde{M}} M \, .$$

Now if X and Y are vector fields on M, then we can extend them locally near M in $\tilde{M}$ to vector fields by the same name. In this situation we have two interesting objects: $\nabla_X Y$, which denotes the application of ∇_X to Y defined by g and $\tilde{\nabla}_X Y$ which is defined by first applying $\tilde{\nabla}$ to the extended vector fields and then restricting the resulting vector fields to a section of $T\tilde{M}|M$. One checks that the result of the latter operation is independent of the choice of the extensions, and we simply denote it by $\tilde{\nabla}_X Y$. It is a section of $T\tilde{M}|M$ which is a direct sum of TM and $N_{\tilde{M}}(M)$,

and therefore splits (orthogonally with respect to $\tilde{g}$) into a sum $t(X,Y)+\Pi(X,Y)$ of sections in the respective bundles. In this way $\Pi=\Pi(X,Y)$ defines a symmetric bilinear pairing on sections of TM with values in N. It is in fact bilinear even over $C^\infty(M)$. Here we have simply defined $t(X,Y)$ as the *tangential* part of $\tilde{\nabla}_X Y$, i.e., it is a vector field on M. The pairing Π is called the second fundamental form defined by the embedding $M\hookrightarrow\tilde{M}$.

Gauss' Formula states that $t(X,Y)=\nabla_X Y$! This result is often simply stated as

$$\tilde{\nabla}_X Y=\nabla_X Y+\Pi(X,Y)\,.$$

It is of course of great interest to understand the relation between the Riemannian curvatures defined in the two different ways. For this we take four fields on M and define $\mathrm{Rm}(X,Y,Z,W)=\langle R(X,Y)Z,W\rangle$ where R is the curvature defined by the Levi-Civita connection of the metric g. We let $\widetilde{\mathrm{Rm}}$ be defined by $\tilde{\nabla}$, i.e., by extending, applying the connection of the ambient space, restricting back to M and then taking the metric product. In this case the resulting formula is not as simple as the above mentioned Gauss formula, but it is a formula:

$$\widetilde{\mathrm{Rm}}(X,Y,Z,W)=R(X,Y,Z,W)-\langle\Pi(X,W),\Pi(Y,Z)\rangle+\langle\Pi(X,Z),\Pi(Y,W)\rangle\,.$$

This is called the *Gauss Equation*.

Gaussian curvature of a surface

Originally Gauss considered 2-dimensional submanifolds (surfaces) in $\mathbb{R}^3$. He defined curvature *extrinsically*, i.e., using the metric/connection of the ambient space (see, e.g., Lee). Given $p\in M$, this is a number $K=K(p)$ which is called the Gaussian curvature of M at p. In the proof of his Theorema Egregium, taking a frame (X,Y), it is shown that

$$K=K(X,Y)=\frac{R(X,Y,Y,X)}{|X\wedge Y|^2} \tag{1}$$

where scaling by the square $|X\wedge Y|^2$ of the area of the parallelogram spanned by X and Y is necessary for K to be an invariant of T_pM which doesn't depend on the frame. Since the RHS of this equation does not depend on the embeddding $M\hookrightarrow\mathbb{R}^3$, this shows that Gauss' extrinsically defined curvature K is actually an *intrinsic* invariant of M. As a result, nowadays one simply defines the Gaussian curvature at a point p in a 2-dimensional Riemannian manifold (M,g) as the RHS of (1).

The exponential map and sectional curvature

Given a point p in a Riemannian manifold (M,g) and a 2-plane π in T_pM there is a canonically determined (local) surface $S(\pi)$ containing p with $T_pS(\pi)=\pi$. The *sectional curvature* $K(\pi)$ of M at p with respect to π is defined to be the Gaussian curvature $K(\pi)$ at p of $S(\pi)$. Hence, sectional curvature at p is a map $K:Gr_2(T_pM)\to\mathbb{R}$ from the Grassmannian of 2-planes. The local surfaces $S(\pi)$ are defined by the exponential map $exp_p:B_r(p)\to M$ which exists on a sufficiently small ball $B_r=\{v\in T_pM:\|v\|^2<r^2\}$ in T_pM. It is defined as follows.

The Euler-Lagrange ODE is a local group action on TM. Let $v_p \in T_pM$ and $\Phi(t)(v_p)$ be its (local) orbit which corresponds to integrating the ODE with the initial condition (p, v_p). Since B_r is compact, for ε sufficiently small there exists $r > 0$ so that $\Phi(t)(v)$ exists for $t \leq 1 + \varepsilon$ for all $v \in B_r$. For $v \in B_r$ one defines $\exp_p(v) := \pi(\Phi(1)(v))$ where $\pi : TM \to M$ is the standard projection. Now the geodesic $\gamma(t) = \exp_p(tv)$, which is defined for $0 \leq t \leq 1$, may be complicated, e.g., with self-intersection. But the differential of $\exp_p$ at $0 \in T_pM$ is the identity, and therefore we may shrink r so that $\exp_p : B_r \to M$ is a diffeomorphism onto an open neighborhood $U = U(p)$. In particular, the inverse of $\exp_p$ on U defines special (normal) coordinates, and in these coordinates the geodesics emanating from p are just straight segments. Of course the geodesics emanating from other points in U are in general not straight lines in these coordinates.

Now every plane $\pi \in Gr_2(T_pM)$ defines a local surface $S(\pi) := \exp_p(\pi \cap B_r)$ in M through p. One defines the *sectional curvature* $K(\pi)$ to be the Gaussian curvature of $S(\pi)$ with respect to its induced metric. However, one can show that the second fundamental form Π of $S(\pi)$ vanishes at p. In particular $\widetilde{\mathrm{Rm}} = \mathrm{Rm}$ at p and consequently $\tilde{K}(\pi) = K(\pi)$. It should be underlined that the full curvature tensor R can be recovered from the sectional curvature and therefore sectional curvature may provide a better geometric understanding of R.

Ricci curvature

The Ricci curvature $\mathrm{Ric} : TM \times TM \to \mathbb{R}$ is the symmetric bilinear form defined by $R(\xi, \eta)$ being the trace of the mapping $\zeta \mapsto R(\zeta, \xi)\eta$. For such a form it is enough to know $R(\xi, \xi)$ for ξ a vector of unit length. In fact, if such a vector $\xi = E_1$ is completed to an orthonormal basis $(E_1, E_2, \ldots, E_n)$, then $\mathrm{Ric}(\xi, \xi) = \sum_{j \geq 2} K(\xi, E_j)$. One says that the Ricci curvature $\mathrm{Ric}(\xi, \xi)$ is the average of the sectional curvatures of the planes which contain the line spanned by ξ. As we will see below, Ricci-flat manifolds play an interesting role in the the classification of manifolds with special holonomy.

Generalities on holonomy

If a vector bundle $E \to M$ over a (connected) manifold is equipped with a connection D and γ is a piecewise smooth curve beginning and ending at a base point $p \in M$, then parallel transport along γ defines a linear transformation $h(\gamma) \in GL(T_pM)$. The holonomy group $\mathrm{Hol}_p = \mathrm{Hol}_p(D)$ is the group of all such transformations. The dependency of Hol_p on p is trivial, and therefore we may fix a base point in the discussion and simply write $\mathrm{Hol} = \mathrm{Hol}(M)$. The restricted holonomy, denoted by $H^0(M)$, consists of those elements $h(\gamma)$ of the holonomy group which are defined by curves are homotopic to zero. If $h = h(\gamma)$ is an arbitrary element of $\mathrm{Hol}(M)$ and $h_t = h(\gamma_t)$ is a curve in $\mathrm{Hol}^0(M)$ defined by a homotopy of γ_1 to the constant loop, then, hh_th^{-1} corresponds to the concatination $\gamma\gamma_t\gamma^{-1}$ and it follows that $hh_1h^{-1} \in \mathrm{Hol}^0(M)$. In other words, $\mathrm{Hol}^0(M)$ is a normal subgroup of $\mathrm{Hol}(M)$ and there is a canonically defined surjective homomorphism $\pi_1(M) \to \mathrm{Hol}(M)/\mathrm{Hol}^0(M)$. In particular, this shows that $\mathrm{Hol}(M)/\mathrm{Hol}^0(M)$ is countable.

It is a basic result which is related to Hilbert's 5th problem and which was proved in its final form in by M. Goto (Proceedings of the AMS, v.20, no.1, (1969)157–162) that a path connected subgroup H of a Lie group G is a Lie subgroup. This means that $H = \langle \exp(\mathfrak{h}) \rangle$ for a Lie subalgebra $\mathfrak{h}$ of $\mathfrak{g}$. Since $\mathrm{Hol}^0(M)$ is contained in the Lie group $GL(T_pM)$ and is by definition path connected, it follows that it is a Lie subgroup of $GL(T_pM)$.

Let us now restrict our attention to the Riemannian case (some results hold for pseudo-Riemannian manifolds) where M is equipped with the associated Levi-Civita connection. In this case $\mathrm{Hol}_p(M)$ is contained in the orthogonal group $O(T_pM, g(p))$ which is compact. This does not mean that $\mathrm{Hol}_p(M)$ is compact, even for compact Riemannian manifolds this may not be the case (see B. Wilking, On compact Riemannian manifolds with noncompact holonomy groups, J.Diff.Geom. 52 (2), 22357 (1999)). On the other hand, using Lie theory Borel and Lichnerowicz (Borel, Armand, Lichnerowicz, Andr, Groupes d'holonomie des varits riemanniennes, Les Comptes rendus de l'Acadmie des sciences, 234 (1952) 1835837) showed that for Riemannian manifolds the connected component $H^0(M)$ is closed in the orthogonal group and is therefore compact.

The *deRahm Decomposition Theorem* is of fundamental importance for the study of the holonomy of Riemannian manifolds. The background for this theorem requires our revisiting the question of splitting, even for a vector space sequence $0 \to V \to W \to Q \to= 0$. Suppose for example that U is a linear operator on W which leaves V invariant. It is immediate that it then induces a linear operator on the quotient Q. The question is if there is a U-splitting of V. In this case this is just a question if there is a U-invariant complement H to V in W. This answer is in general "No"! For example, if $W = \mathbb{R}^2$ equipped with its standard basis (e_1, e_2), $V = \mathbb{C}.e_1$ and U is an operator with matrix

$$\mathrm{Mat}(U) = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix},$$

then there is no U-splitting. In this case U even acts trivially on the quotient, but there is not a U-invariant complement to V!

The same discussion applies to group representations, i.e., to a homomorphism $\rho : G \to GL(W)$ where W is a (finite-dimensional) vector space. One uses an abbreviated notation $g(w) = \rho(g)(w)$ and says that G stabilizes a subspace V whenever for every g in G it follows that $g(v) \in V$ for all $v \in V$. The same discussion as that above applies to representations: If G stabilizes V, then it acts on the quotient $Q = W/V$. However, G may or may not stabilize a complementary subspace H to V. There is one standard situation where the existence of complementary stabile subspaces is guaranteed. This is where W is equipped with a G-invariant (positive-definite) metric b: If V is G-invariant, then $W = V \oplus V^{\perp_b}$ is a G-invariant decomposition. In this case, the representation is said to be *completely reducible*, because repeated application of the above yields a decomposition

$$W = V_1 \oplus \cdots \oplus V_m$$

where each V_i possesses no non-trivial invariant supspaces and which is unique up to order.

In the case where $G = \mathrm{Hol}(M)$ is the holonomy group acting on the tangent space T_pM, the Riemann metric $g(p)$ is preserved by G and thus the holonomy representation of $\mathrm{Hol}(M)$ on T_pM is completely reducible. One says that the holonomy representation on M is irreducible, or simply that M is irreducible, if T_pM possesses no non-trivial Hol-invariant subspaces. The deRham decomposition theorem provides a method of reducing to the situation where M is irreducible. If $T_pM = V_1 \oplus \ldots \oplus V_m$ is a decomposition into irreducible subspaces, then parallel transport yields a corresponding decomposition $TM = E_1 \oplus \ldots \oplus E_m$ of subbundles. In fact these subbundles are integrable in the sense of Frobenius: at least locally $M = M_1 \times \ldots \times M_m$ with $E_i = TM_i$, $i = 1, \ldots, m$. If $\pi_1(M) = 1$, e.g., if we move to the universal cover $\tilde{M}$ of the original manifold, then this splitting is global! Thus in this sense we have a global $\mathrm{Hol}^0(M)$-splitting. Quite often this only reduces questions to studying phenomena related to actions of discrete groups Γ where $M = \tilde{M}/\Gamma$. But nevertheless it is a major reduction.

Cartan's Dichotomy

In the early part of the 20th century E. Cartan proposed guiding the study of Riemannian manifolds by dividing them into two classes depending on whether curvature is invariant by parallel transform, i.e., $DR = 0$, or not. He also introduced holonomy with this project in mind. Many years later Berger (Les espaces symetriques noncompacts, Ann.Sci.Ecol.Norm.Sup. 74, 85-177 (1957)) and then Simons (On transitivity of holonomy systems, Ann. Math. 76, 213-234 (1962)) proved the following fundamental result.

Theorem 1. *If M is an irreducible Riemannian manifold and does not satisfy $DR = 0$, then its holonomy group $\mathrm{Hol}^0_p(M)$ acts transitively on the unit sphere in T_pM.*

Of course the orthogonal group and the special orthogonal group act transitively on the sphere and we refer to this as the generic case. But if $\mathrm{Hol}^0(M)$ is not one of these groups, there are very few classes of examples. In other words, Berger's Theorem really focuses the study. Here is a list of these special holonomy groups, i.e., those other than the generic groups, and the corresponding names of the manifolds.

Hol	dim(M)	Metric
U(n)	2n	Kähler
SU(n)	2n	Calabi-Yau, Kähler
Sp(n)	4n	Hyperkähler
Sp(n).U(1)	4n	Quaternionic-Kähler
G_2	7	G_2-manifold
$Spin(7)$	8	$Spin(7)$-manifold

It should be underlined that except for the general Kähler manifold, and the Quaternionic-Kähler manifolds which are Einstein, all manifolds in the list are Ricci flat. The Einstein condition is that $\mathrm{Ric} = kg$ where k is a constant. Of course $k = 0$ means Ricci-flat. It should also be stressed that it is often extremely difficult to

find or prove the existence of a metric in these cases. For example in work of Aubin and of Yau, the latter receiving a Fields Medal for his work in this area, Ricci-flat Calabi-Yau metrics are constructed on certain important 4-dimensional (two complex dimensions) manifold. This work is extremely technical and furthermore the resulting metrics exist (by limiting procedures) but have never been explicitly seen.

Manifolds satisfying $DR = 0$

Cartan showed that manifolds M which satisfy his conditions $DR = 0$ are *locally summetric*. This means that the universal cover $\tilde{M}$ of M is a globally symmetric space. A Riemannian manifold (M, g) is globally symmetric if, given a point $p \in M$ there exists a globally defined isometry σ in $Iso(M)$ which fixes p and satisfies $\sigma_*^2 = -Id$. It follows that $M = G/K$ is the homogeneous space of a connected Lie group G with compact isotropy group K. Of course G and K are of a special type which have in the meantime been classified in detail. Perhaps one of the main points is that the reflection σ induces a splitting $\mathfrak{g} = \mathfrak{k} \oplus \mathfrak{p}$ where $\mathfrak{p}$ is identified with the tangent space T_pM at the base point p being considered, and the geodesics emanating from p are simply the orbits $g_\xi(t).p$ of the 1-parameter groups g_ξ with $\xi \in \mathfrak{p}$. This gives us a major additional tool for describing the geodesics. Of course this is not quite the panacea one would hope for, because the locally symmetric spaces are of the form M/Γ where M is globally symmetric and Γ is a discrete subgroup in G. So we have shifted the difficulties to those involving discrete groups. Our last block of lectures (May 16/17) will be devoted to these locally and global symmetric spaces.